

Albany RR300 Clean

The certified clean room high speed door

Albany RR300 Clean

- Cleanroom suitability certified by Fraunhofer Institute
- Fulfils international standards and guidelines for cleanroom applications in accordance with DIN EN ISO 14644-1 and GMP
- ISO Class 5, GMP Class C
- Low particle emission
- Low air exchange thanks to fast opening and closing speed



DOOR DESIGN

The entire door (side frame, motor and roll cover, and bottom profile) is made from stainless steel. The smooth surface facilitates easy cleaning and avoids the accumulation of particle deposits.



NARROW SIDE COLUMNS / MINIMAL GAP

The dimensions of the door's side columns are kept to a minimum and allow the door to be installed in confined areas. The extremely small gap of the door curtain guide ensures minimal pressure loss.



CABLE GUIDE SEALING

To prevent pressure losses, the cable guide is integrated and access openings inside the door are sealed.



DOOR CURTAIN

The door curtain consists of PVC with white fabric reinforcement strips. Fabric reinforcement strips in a wide variety of RAL colors are available as option.



BOTTOM PROFILE

There is only a small gap between the bottom profile and the side columns. This minimises the pressure loss when the door is closed.



SAFETY FEATURES

The door is equipped with an electrical safety contact edge. A contactless pre-running safety photocell is available as option. In addition the door is equipped with a door line photocell which prevents the door from closing when an object is in the doorway.



INTEGRATED CABLE GUIDE

The cables are located in the side columns to provide a smooth surface and avoid surface irregularities that increase particle turbulence.

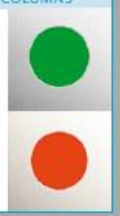
The supply cables of the pre-running safety photocell and teh electrical safety contact edge are guided reliably by internal cable carrier chain.



INTEGRATED TRAFFIC LIGHT IN THE SIDE COLUMNS

It is possible to signal the movement of the door via a traffic light system which can be inserted into either both side columns or just one side column.

A yellow flashing light, a stationary red/green light is available or a traffic light in separate stainless steel housing could be installed opposite. A red/green traffic light with recognition of direction is possible.



TOP ROLL COVER / MOTOR COVER

The door is equipped with top roll cover and motor cover as standard.



FREQUENCY CONVERTER CONTROL SYSTEM

The Albany MCC frequency converter control system and the optional UPS battery are located in a compact sealed compartment under the top roll cover.



MOTOR POSITION

The motor can be fitted either on the right or on the left hand side.



CONTROL UNIT AND MAIN SWITCH

The display unit with its user-friendly foil keypad and graphic display and the main switch are integrated in the side column.



EGRESS

Self opening is available as an option. The UPS battery supplies power to the door control system in case of a power failure. UPS error and diagnostic messages are indicated by LEDs.



CONTACTLESS ACTUATOR

An optional contactless actuator is available for fast, hygienic opening.



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Albany RR300 Clean

DW: 1000-3500 mm

DH: 1000 -3500 mm

Side frames: Stainless steel

Control: MCC – Integrated underneath covers

Opening/Closing speed: 2,0/1,0 m/s

Low leakage (max. 20m³/h at a pressure difference of 25 Pa and door size of 3 m²)

Suitable for cleanrooms with pressure differences up to 50 Pa

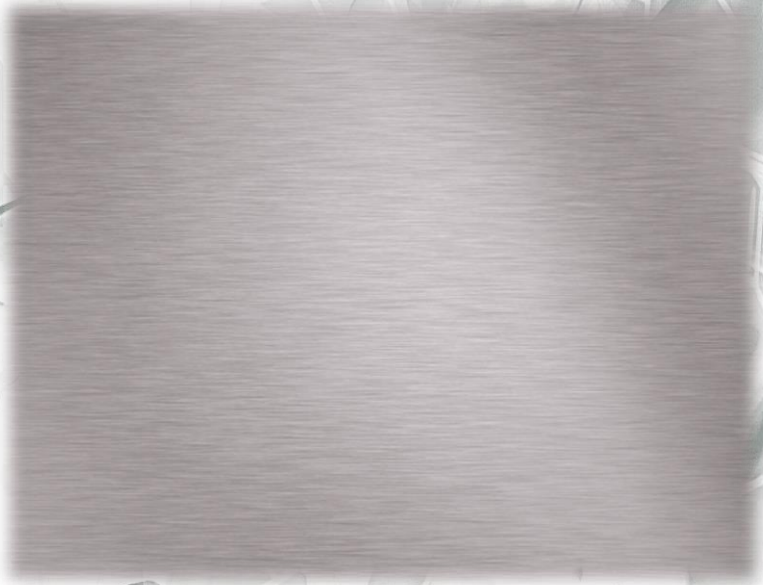
Isolation in accordance with DIN EN **ISO** 14644-1 **Class 5**

Design in accordance with **GMP Class C**

2 Safeties: Contact edge/Pre-running photocells and door-line photocell

Motor and top roll cover included

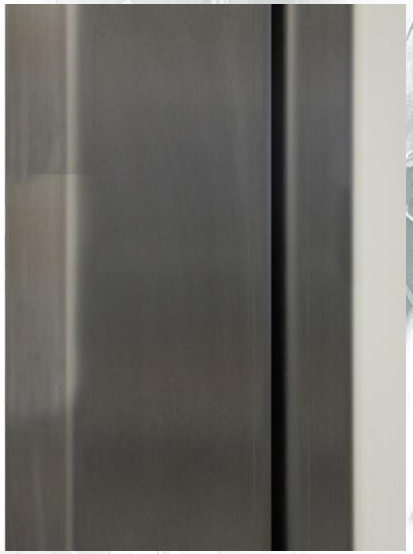
IPA Fraunhofer TESTED® DEVICE
Albany Door Systems
RapidRoll Clean
Report No. AL 1003-514



DOOR MATERIAL

The entire door (side frame, motor and barrel cover, bottom profile) is made of stainless steel with a minimum surface roughness. The extremely smooth surface facilitates easy cleaning and prevents the accumulation of particle deposits.





Narrow side columns / minimum gap

The ultra tight gap of only 5 mm fits closely to the curtain minimising pressure loss and particle emmissions due to friction. The slim columns allow the door to be installed in areas tight on valuable production space and provide for a sleek appealing design.



Side Frame

- Compact design
- Including factory sealings
- Pre-assembled





Bottom profile

Is guided by special long life sliders designed to only leave a small gap between the bottom profile and the side columns. The elongated sealing lip is preformed in the correct position to fit your pressurised area.





Bottom Profile

- Long life time sliders
- Pre-formed lip





Integrated cabling

Cables are located inside the side columns to guarantee a flush external surface and and avoid the spread of partical dust due to cable movement.

The supply cables of the contact strip are guided reliably by an internal cable chain.









Internal Sealings

All openings are internally sealed off to prevent pressure loss and insure the sleek but effective design. The main power supply membrane is the only visible cable sealing.



Stainless steel motor and barrel covers 45 ° graded.





Frequency converter control

- MCC Vector Control frequency converter
- Opening speed up to 2 m/s
- Optional Uninterruptible Power Supply (UPS) integrated underneath barrel cover
- Display unit and main switch located in side column



MCC Expansion box

Expansion box with interface card integrated underneath top roll cover

(Required for 230 V traffic light, potential free contact and interlock with other control systems)







Door curtain - Default

Transparent PVC with white (RAL 9010) silicone free reinforcement stripes



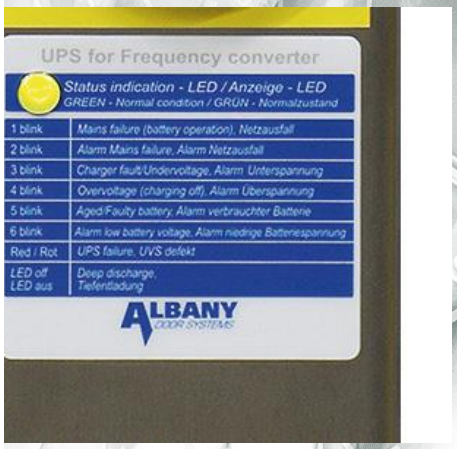


Door curtain - Optional

Optional antistatic door blade „NOMATEX“, grey RAL7038, 2,05 mm thickness, FDA-approved







Diagnostic Display Unit / Main switch Self Opening

The user-friendly foil keypad with integrated graphic display and the main switch are integrated in the side column.

Optional UPS battery supplies power to the door in case of a power failure. LED signals battery status.





Safety features

The door is equipped with an electrical safety contact strip. A contactless pre-running safety photocell is optionally available instead of the contact strip.
Both of these safety devices are also supported by a stationary
safety photocell, which prevents the door from closing when objects are in the way.





Contactless impulse activator *”MAGIC SWITCH”*

A contactless open switch is optionally available for fast, hygienic opening.



Difference between a Conventional door and a RR300 Clean

Conventional	RR300 Clean
Side columns: Composite material or low grade SS or powder coat / rough curtain (or windbars)	**Side columns:** Smooth SS Surface / smooth PVC has excellent cleaning qualities (no windbars)
Leakage rate (m^3/h at 25 Pa) Up to 200 m^3/h	**Leakage rate (m^3/h at 25 Pa)** Institute Test of 20 m^3/h

Integrated traffic light in side columns (optionally)

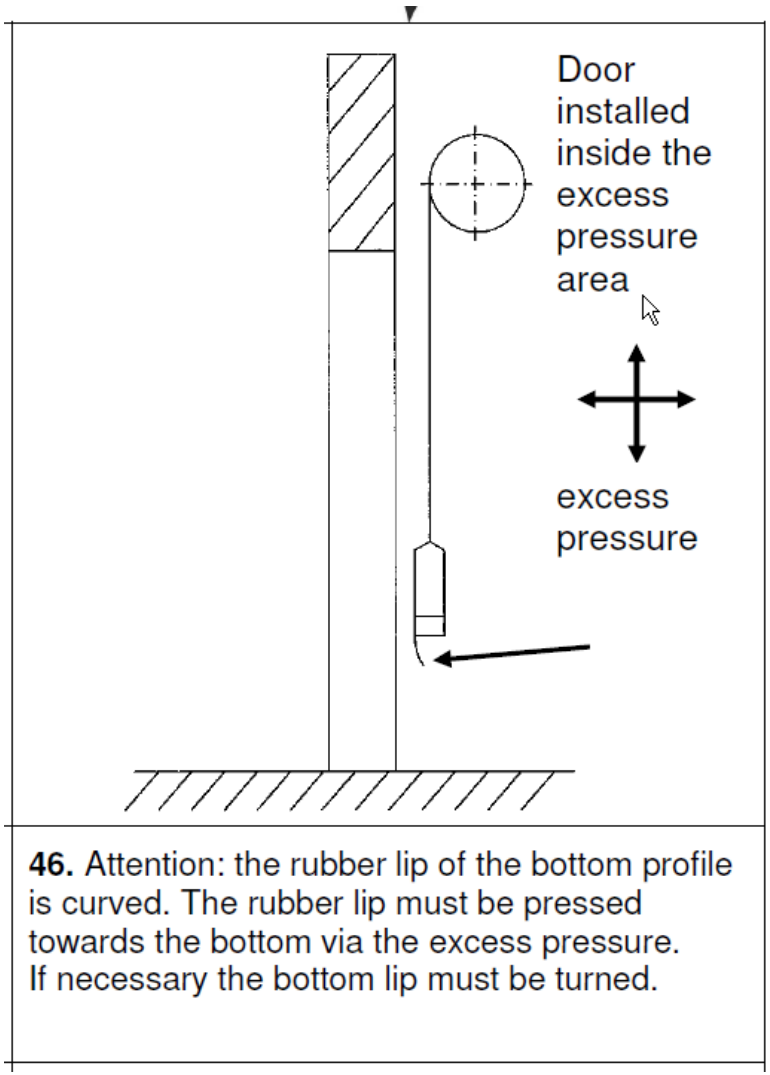
- Movement of the door can be signaled via a traffic light system
- Inserted into one side column or both sides
- Yellow flashing light
- Stationary red/green light
- Traffic light in separate stainless steel housing to be installed on opposite side



Sealing function

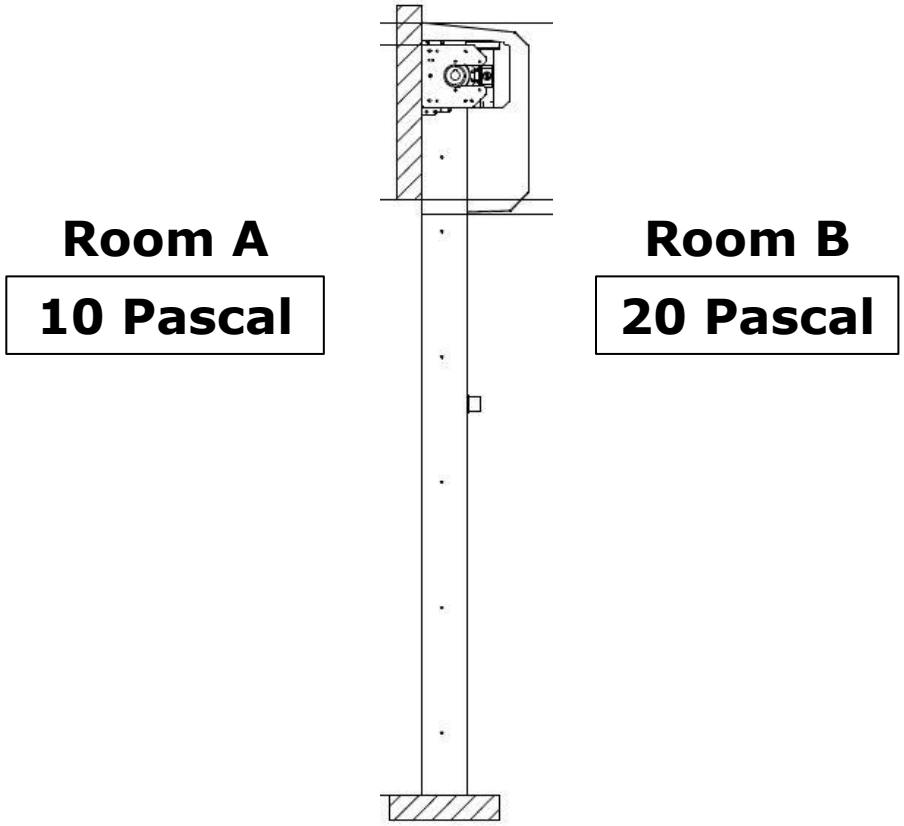
- Sealing value is the same from both sides
- Lip must be curved in the correct direction (from Factory)





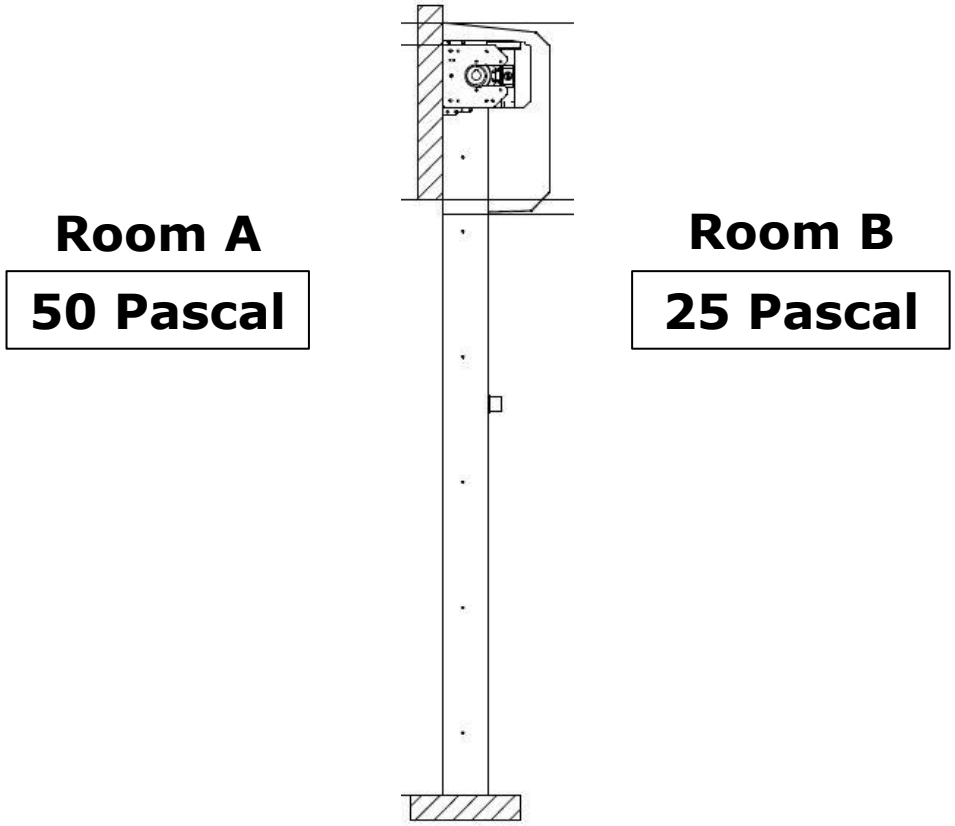
46. Attention: the rubber lip of the bottom profile is curved. The rubber lip must be pressed towards the bottom via the excess pressure. If necessary the bottom lip must be turned.

Installation position of RR300 Clean - Inside or outside overpressured area?



<u>Example 1</u>: Installation inside overpressured area

Installation position of RR300 Clean - Inside or outside overpressured area?



Example 2: Installation outside overpressured area

Interlock situation



Case study

Albany equips cleanroom airlocks of Merck in Puerto Rico



The production of medication requires surroundings free of particles because already smallest particles could contaminate the pharmaceutical products.

In order to ensure such surroundings the *Merck* pharmaceutical group implemented several cleanrooms at its production plant in Puerto Rico.

The engineering company *Daldrop + Dr. Eng. Huber*, based in Germany was responsible for planning and implementation.

Eight clean room doors of the type **Rapid Clean** have been integrated to control the access to the material locks.

This high performance door conforms to the Good Manufacturing Practice (GMP) and has been certified (ISO) by the Fraunhofer Institute regarding its ability for cleanrooms.

Albany RR300 Plus Alu

- Compact, lightweight high speed door for internal use
- Additional equipment available for application in emergency exit areas

ALBANY RR300 PLUS ALU SEALED

Amongst others the curtain guidance in the sideframe is reduced in order to achieve an optimum of sealing and a minimum loss of pressure.

It is suitable for the application in cleanrooms. The door can be installed inside or outside the positive pressure region.

The door can be applied up to ISO class 7.

ALBANY RR300 PLUS ALU F+R

By means of special additions the Albany RR300 Plus Alu can be transferred into a high performance door for emergency exits and escape routes (up to a maximum size of 4000 x 4000 mm).

In this case the door is equipped with several monitoring devices, e.g. radar motion detector, automatic self-opening and regular self-testing.

The model Albany RR300 Plus Alu Sealed is not suitable as emergency exit.

MOTOR COVER AND TOP ROLL COVER

Standard: without cover.
(for standard door Albany RR300 Plus Alu)

Scope of delivery includes drive unit and top roll cover
(for door Albany RR300 Plus Alu with additional sealing)

Scope of delivery includes top roll cover
(for the emergency exit and escape route version)

Optional: motor cover is available in plastic RAL 7 040, stainless steel or steel powder coated RAL 9006 (for standard door Albany RR300 Plus Alu)

Optional: top roll cover is available in aluminum.
(for standard door Albany RR300 Plus Alu)

DOOR DESIGN

The door (side frame, top roll cover and bottom profile) is made from aluminium.

DOOR CURTAIN

Standard: door is made of clear PVC with blue/red reinforcement stripes.

Optional: door curtain includes RollTex® or NomaTex® curtain with / without windows or vision panels or PVC with silicon free reinforcement stripes.



ACTUATORS

Individual applications can include additional safety devices. Actuators such as infrared, radar detectors, push buttons, vehicle detector loops or radio transmitters can be added to the control system. Please contact us for additional information for your specific needs.

KNOCK-OUT CAPABILITY

The Albany RR300 Plus Alu can be equipped with a knock-out capability option. In the event of a collision the bottom profile breaks-away, causing a sensor to stop the door. The door can be easily reset by hand.



SAFETY FEATURES

The door has an electrical safety contact edge. Standard in-line photocell prevents the door from closing when an object is in the doorway. An optional integrated light curtain is available for doors with a minimum height of 1975 mm.



INTEGRATED CABLE CHAIN

As standard the door is equipped with an internal cable routing to the bottom profil.

MOTOR POSITION

Gear motor, available left or right hand side.

CONTROL BOX

As standard the door is made of a relay control system MCS-R.

According to the operation conditions optional a contactor control system ACS 50 and a frequency converter control system MCS-FC and MCC are available. With the basic control system MCS-R and MCS-FC a convenient operation with high flexibility, fast start-up and easy handling of the door is allowed. The powerful MCC frequency converter control system with Torque Vectoring of the drive unit, connected to the wide range of functions, enables nearly an unlimited use of the door. The MCC guarantees a soft start, optimal increase in speed, maximum speeds and a long service life of the door.

Figure shows MCC control system.





MANUAL OR AUTOMATIC EGRESS

Manual or automatic egress opens the door up to 2.0 m in the event of a power failure.



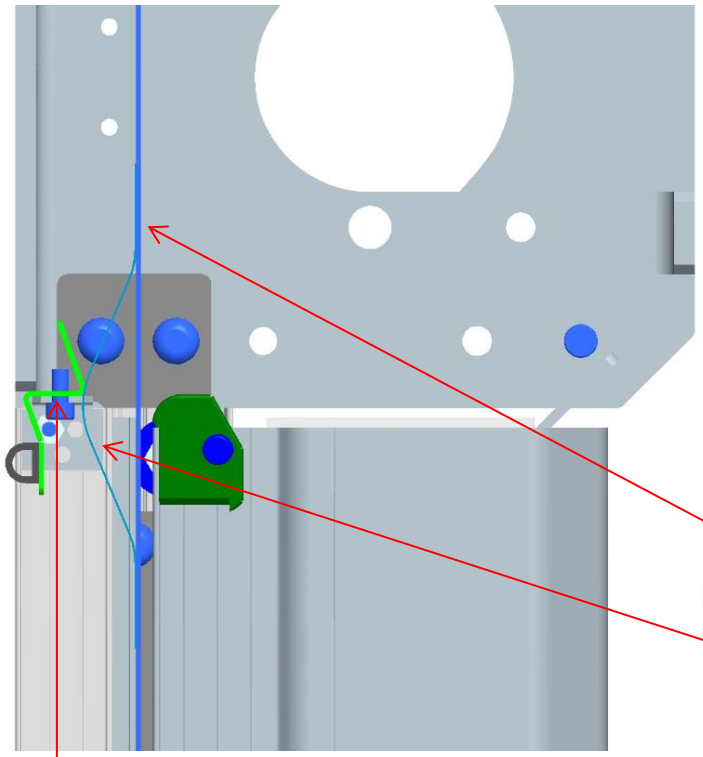
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RR 300+ Sealed - Low Spec Clean Door

- DW: 1000-3000 mm // DH: 1000-3000 mm
- Wind class: 0 // Max. pressure difference 70 Pa
- Air permeability: 65 m³/h at pressure difference 25 Pa (1500x2000 mm)
- Side frames: Aluminium with bottom plates
- Bottom profile: Anodized aluminium (Mini version),without crash
- Electrical safety edge
- Speed: Opening 2,1 m/s – Closing 0,8 m/s
- Mark up in option price list of RR300+
- Application:
  - -> Clean Rooms without ISO certification
  - -> other pressured areas with need of low leakage



RR300 Plus Sealed – Low Spec Clean Door



Specification:

- RR300 Plus with ALU sides (not Inox)
- Max.: 3000 x 3000 mm
- Guide rail: 6 mm gap instead of 12 mm
- Sealed head and foot plates
- Header sealing as on RRCR
- Leakage vs RR300 Clean: 3 times higher
- Leakage vs RR300+: 8-10 times lower
- Same rate front / back
- F:Bag pushes against lintel profile
- B:Curtain against rounded barrel egde


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Lilly
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